12TH CHEMISTRY · CHAPTER 18 · FULL CHAPTER MCQS

Transition Metals

107 topic-wise MCQs from the complete chapter with answers and reasons

18.1 FIRST ROW TRANSITION ELEMENTS (3d)

1. Transition elements are found in which block(s) of the periodic table?

(a) s and p
(b) d and f
(c) p only
(d) s only

Answer: (b) d and f — Stated in the introduction.

2. A transition element has partially filled:

(a) d-orbitals
(b) s-orbitals
(c) p-orbitals
(d) Valence shell only

Answer: (a) d-orbitals — In any of its stable oxidation states.

3. Which d-block element is NOT a transition element?

(a) Iron
(b) Copper
(c) Zinc
(d) Nickel

Answer: (c) Zinc — Zn^{2+} has a complete 3d subshell.

4. The configuration of Zn^{2+} is:

(a) [Ar] $3d^8\ 4s^2$
(b) [Ar] $3d^{10}\ 4s^2$
(c) [Ar] $3d^9\ 4s^1$
(d) [Ar] $3d^{10}\ 4s^0$

Answer: (d) [Ar] $3d^{10}\ 4s^0$ — The 4s electrons are lost.

5. The electronic configuration of chromium is:

(a) [Ar] $3d^4\ 4s^2$
(b) [Ar] $3d^6\ 4s^0$
(c) [Ar] $3d^5\ 4s^1$
(d) [Ar] $3d^3\ 4s^2$

Answer: (c) [Ar] $3d^5\ 4s^1$ — Half-filled d-subshell is more stable.

6. The electronic configuration of copper is:

(a) [Ar] $3d^{10}\ 4s^1$
(b) [Ar] $3d^9\ 4s^2$
(c) [Ar] $3d^8\ 4s^2$
(d) [Ar] $3d^{10}\ 4s^2$

Answer: (a) [Ar] $3d^{10}\ 4s^1$ — Completely filled d-subshell is more stable.

studysapiens.com

7. Which two elements do not follow the Aufbau principle in the 3d series?

(a) Ti and V
(b) Mn and Ni
(c) Fe and Co
(d) Cr and Cu

Answer: (d) Cr and Cu — Section 18.1.1.

8. The configuration of Fe^{3+} is:

(a) [Ar] $3d^6$ $4s^0$
(b) [Ar] $3d^5$ $4s^0$
(c) [Ar] $3d^3$ $4s^2$
(d) [Ar] $3d^4$ $4s^1$

Answer: (b) [Ar] $3d^5$ $4s^0$ — Table 18.1.

9. The configuration of Cu^{2+} is:

(a) [Ar] $3d^9$ $4s^0$
(b) [Ar] $3d^{10}$ $4s^0$
(c) [Ar] $3d^8$ $4s^1$
(d) [Ar] $3d^7$ $4s^2$

Answer: (a) [Ar] $3d^9$ $4s^0$ — Table 18.1.

10. The number of 3d electrons in Ti^{3+} is:

(a) 0
(b) 2
(c) 1
(d) 3

Answer: (c) 1 — [Ar] $3d^1$ $4s^0$.

18.2 PROPERTIES OF TRANSITION ELEMENTS

11. Which 3d metal is reddish brown?

(a) Iron
(b) Titanium
(c) Nickel
(d) Copper

Answer: (d) Copper — The others are silvery or grey.

12. Transition metals are paramagnetic because of:

(a) Paired electrons
(b) Filled 4s subshell
(c) Unpaired d electrons
(d) High density

Answer: (c) Unpaired d electrons — Section 18.2.1.

13. Across the 3d row from Ti to Cu, density generally:

(a) Decreases
(b) Increases
(c) Stays constant
(d) Becomes zero

Answer: (b) Increases — Nuclear charge builds up and size shrinks.

studysapiens.com

14. "Malleable" means the metal can be:

(a) Hammered into thin sheets (b) Drawn into wires
(c) Melted easily (d) Magnetised

Answer: (a) Hammered into thin sheets — Ductile means drawn into wires.

15. The 3d metal with the highest melting point (Figure 18.1) is:

(a) Titanium (b) Iron
(c) Vanadium (d) Copper

Answer: (c) Vanadium — 1910 °C.

16. The maximum oxidation state of manganese is:

(a) +5 (b) +4
(c) +6 (d) +7

Answer: (d) +7 — The highest in the 3d series.

17. The common oxidation state obtained by losing only the 4s electrons is:

(a) +1 (b) +2
(c) +3 (d) +4

Answer: (b) +2 — Section 18.2.3.

18. The highest oxidation state of chromium is:

(a) +6 (b) +3
(c) +7 (d) +5

Answer: (a) +6 — Table 18.2.

19. Variable oxidation states arise because 3d and 4s subshells have:

(a) Very different energy (b) No electrons
(c) The same shape (d) Comparable energy

Answer: (d) Comparable energy — Both sets of electrons are used.

20. The catalyst used in the Haber process is:

(a) Nickel (b) Iron
(c) Copper (d) Vanadium

Answer: (b) Iron — A heterogeneous catalyst.

21. Nickel salts are used in the preparation of acrylic acid from:

(a) Ethene (b) Benzene
(c) Acetylene (d) Methane

Answer: (c) Acetylene — Involves Ni(0) and Ni(II).

studysapiens.com

22. In heterogeneous catalysis, reactants attach to the catalyst surface and form:

(a) A temporary activated complex
(b) A permanent compound
(c) A precipitate
(d) An alloy

Answer: (a) A temporary activated complex — It lowers the activation energy.

23. A catalyst provides a pathway with:

(a) Higher activation energy (b) Higher enthalpy change
(c) Lower activation energy (d) No intermediate

Answer: (c) Lower activation energy — Section 18.2.4.

18.3 TRANSITION METAL COMPLEXES

24. In complex formation the transition metal ion acts as a:

(a) Lewis base (b) Lewis acid
(c) Brønsted base (d) Reducing agent

Answer: (b) Lewis acid — It accepts electron pairs.

25. The bond between a ligand and the central metal ion is:

(a) Ionic (b) Metallic
(c) Dative (coordinate covalent) (d) Hydrogen bond

Answer: (c) Dative (coordinate covalent) — The ligand donates both electrons.

26. The word “ligand” comes from the Latin “ligare”, meaning:

(a) To give (b) To colour
(c) To break (d) To bind

Answer: (d) To bind — Section 18.3.

27. The name of the ligand CN^- is:

(a) Cyano (b) Cyanide
(c) Carbonyl (d) Cyanium

Answer: (a) Cyano — Anionic ligand names end in “-o”.

28. The name of NH_3 as a ligand is:

(a) Ammonia (b) Ammine
(c) Amino (d) Ammonium

Answer: (b) Ammine — A special name.

29. The name of CO as a ligand is:

(a) Carbonate (b) Carbon
(c) Carboxyl (d) Carbonyl

Answer: (d) Carbonyl — A neutral ligand.

30. Which is a cationic ligand?

(a) NO^+ (b) Cl^-
(c) H_2O (d) CO

Answer: (a) NO^+ — Nitrosonium.

31. Which is a monodentate ligand?

(a) EDTA (b) 1,2-diaminoethane
(c) NH_3 (d) Ethanedioate ion

Answer: (c) NH_3 — It donates only one pair.

32. Which is a bidentate ligand?

(a) H_2O (b) CN^-
(c) EDTA (d) 1,2-diaminoethane

Answer: (d) 1,2-diaminoethane — Each N donates a lone pair.

33. The ethanedioate ion donates electron pairs through two:

(a) Carbon atoms (b) Oxygen atoms
(c) Nitrogen atoms (d) Hydrogen atoms

Answer: (b) Oxygen atoms — One from each carboxyl group.

34. EDTA is a:

(a) Hexadentate ligand (b) Bidentate ligand
(c) Monodentate ligand (d) Tetradentate ligand

Answer: (a) Hexadentate ligand — Six donor atoms.

35. The donor atoms of EDTA are:

(a) 6 oxygen (b) 2 oxygen and 4 nitrogen
(c) 4 oxygen and 2 nitrogen (d) 6 nitrogen

Answer: (c) 4 oxygen and 2 nitrogen — Section 18.3.

36. The abbreviation *en* stands for:

(a) 1,2-diaminoethane (b) Ethanedioate
(c) EDTA (d) Ethene

Answer: (a) 1,2-diaminoethane — Also called ethylenediamine.

37. The coordination sphere is usually written inside:

(a) Round brackets (b) Square brackets
(c) Braces (d) No brackets

Answer: (b) Square brackets — [Complex].

38. The charge on iron in $[Fe(CN)_6]^{4-}$ is:

(a) +3 (b) +4
(c) +2 (d) –4

Answer: (c) +2 — x + 6(–1) = –4.

studysapiens.com

18.4 PREDICTING FORMULA AND CHARGE

39. The charge on the complex of Co^{3+} with six NH_3 ligands is:

(a) +2 (b) –3
(c) 0 (d) +3

Answer: (d) +3 — (+3) + 6(0).

40. The oxidation state of nickel in $[Ni(CO)_4]$ is:

(a) +2 (b) 0
(c) +4 (d) +1

Answer: (b) 0 — CO is neutral and the complex is neutral.

41. $[Ni(CO)_4]$ is a:

(a) Neutral complex (b) Cationic complex
(c) Anionic complex (d) Simple salt

Answer: (a) Neutral complex — Table 18.3.

42. A complex of Fe^{2+} with six CN^- ligands has the charge:

(a) +4 (b) –6
(c) –4 (d) +2

Answer: (c) –4 — (+2) + 6(–1).

43. In the general formula $[M(L)_x]^{\pm y}$, x is fixed by the:

(a) Coordination number (b) Atomic number
(c) Mass number (d) Period number

Answer: (a) Coordination number — The number of ligands.

18.5 GEOMETRY OF COMPLEXES

44. The bond angle in a linear complex is:

(a) 90° (b) 120°
(c) 109.5° (d) 180°

Answer: (d) 180° — Ligands on opposite sides.

45. $[Ag(NH_3)_2]Cl$ contains a complex ion that is:

(a) Tetrahedral (b) Octahedral
(c) Linear (d) Square planar

Answer: (c) Linear — Coordination number 2.

46. Square planar geometry is characteristic of which metal ions?

(a) d^5 (b) d^8
(c) d^{10} (d) d^0

Answer: (b) d^8 — e.g., Ni^{2+} and Pt^{2+}.

studysapiens.com

47. The geometry of $[Ni(CN)_4]^{2-}$ is:

(a) Square planar (b) Tetrahedral
(c) Octahedral (d) Linear

Answer: (a) Square planar — Section 18.5.

48. The geometry of $[CoCl_4]^{2-}$ is:

(a) Square planar (b) Octahedral
(c) Tetrahedral (d) Linear

Answer: (c) Tetrahedral — Bond angle 109.5°.

49. The most common geometry of transition metal complexes is:

(a) Linear (b) Octahedral
(c) Tetrahedral (d) Square planar

Answer: (b) Octahedral — Six ligands.

50. The number of ligands in an octahedral complex is:

(a) 6 (b) 4
(c) 2 (d) 8

Answer: (a) 6 — Along the x, y and z axes.

51. $[Co(en)_3]^{3+}$ is:

(a) Tetrahedral (b) Square planar
(c) Linear (d) Octahedral

Answer: (d) Octahedral — Three bidentate ligands give coordination number 6.

18.6 COLOUR OF COMPLEXES

52. The number of d-orbitals is:

(a) 5 (b) 3
(c) 7 (d) 1

Answer: (a) 5 — Five distinct orbitals.

53. Orbitals of the same energy are called:

(a) Hybrid (b) Split
(c) Non-degenerate (d) Degenerate

Answer: (d) Degenerate — As in an isolated atom.

54. In an octahedral complex, the higher-energy orbitals are:

(a) d_{xy}, d_{yz}, d_{xz}
(b) $d_{x^2-y^2}$, d_{z^2}
(c) d_{xy} only
(d) All five

Answer: (b) $d_{x^2-y^2}$, d_{z^2} — Two higher, three lower.

studysapiens.com

55. In a tetrahedral complex, the number of higher-energy d-orbitals is:

(a) 2 (b) 5
(c) 3 (d) 1

Answer: (c) 3 — The splitting is inverted.

56. Compared with octahedral complexes, splitting in tetrahedral complexes is:

(a) Smaller (b) Larger
(c) Equal (d) Zero

Answer: (a) Smaller — Section 18.6.

57. A complex that absorbs all white light appears:

(a) White (b) Colourless
(c) Red (d) Black

Answer: (d) Black — Section 18.6.1.

58. A complex that absorbs yellow light appears:

(a) Green (b) Violet
(c) Orange (d) Red

Answer: (b) Violet — The complementary colour.

59. The complementary colour of red is:

(a) Blue (b) Violet
(c) Green (d) Yellow

Answer: (c) Green — Opposite on the colour wheel.

60. The complementary colour of orange is:

(a) Green (b) Blue
(c) Violet (d) Red

Answer: (b) Blue — Opposite on the colour wheel.

61. Planck's equation is:

(a) $\Delta E = h\nu$ (b) $E = mc^2$
(c) $PV = nRT$ (d) $E = \frac{1}{2}mv^2$

Answer: (a) $\Delta E = h\nu$ — $\Delta E = h\nu = hc/\lambda$.

62. Which ligand causes the largest d-orbital splitting?

(a) I^- (b) Cl^-
(c) H_2O (d) CN^-

Answer: (d) CN^- — First in the series.

63. Which ligand causes the smallest splitting?

(a) NH_3 (b) OH^-
(c) I^- (d) en

Answer: (c) I^- — Last in the series.

64. A larger ΔE corresponds to absorption of light with:

(a) Shorter wavelength (b) Longer wavelength
(c) Lower frequency (d) No energy

Answer: (a) Shorter wavelength — Higher frequency.

65. The complex $[Ti(H_2O)_6]^{3+}$ is:

(a) Tetrahedral (b) Square planar
(c) Linear (d) Octahedral

Answer: (d) Octahedral — Six ligands.

18.7 LIGAND EXCHANGE REACTIONS

66. The colour of $[Cu(H_2O)_6]^{2+}$ is:

(a) Blue (b) Pink
(c) Yellow (d) Green

Answer: (a) Blue — As in copper sulphate solution.

67. Adding NaOH to $Cu^{2+}_{(aq)}$ gives a:

(a) Yellow solution (b) Dark blue solution
(c) Pink precipitate (d) Light blue precipitate

Answer: (d) Light blue precipitate — $[Cu(H_2O)_4(OH)_2]$.

68. With concentrated ammonia, $[Cu(H_2O)_6]^{2+}$ gives a solution that is:

(a) Yellow (b) Dark blue
(c) Pink (d) Colourless

Answer: (b) Dark blue — $[Cu(NH_3)_4(H_2O)_2]^{2+}$.

69. How many water ligands of $[Cu(H_2O)_6]^{2+}$ are replaced by ammonia?

(a) Two (b) Six
(c) Four (d) One

Answer: (c) Four — Section 18.7.1.

70. The colour of $[CuCl_4]^{2-}$ is:

(a) Blue (b) Violet
(c) Pink (d) Yellow

Answer: (d) Yellow — Formed with conc. HCl.

71. The colour of $[Co(H_2O)_6]^{2+}$ is:

(a) Pink (b) Blue
(c) Yellow (d) Green

Answer: (a) Pink — As in cobalt(II) sulphate solution.

studysapiens.com

72. Cobalt(II) hydroxide precipitate is:

(a) Pink (b) Blue
(c) Yellow (d) White

Answer: (b) Blue — It may turn pink on standing.

73. The colour of $[Co(NH_3)_6]^{2+}$ is:

(a) Blue (b) Pink
(c) Yellow (d) Green

Answer: (c) Yellow — Hexaamminecobalt(II).

74. The colour of $[CoCl_4]^{2-}$ is:

(a) Pink (b) Blue
(c) Yellow (d) Red

Answer: (b) Blue — A tetrahedral complex.

75. Ligand exchange changes the colour because it changes:

(a) The atomic number (b) The metal
(c) The nucleus (d) ΔE

Answer: (d) ΔE — A different frequency is absorbed.

18.8 STABILITY CONSTANT

76. K_{stab} is a type of:

(a) Rate constant (b) Gas constant
(c) Equilibrium constant (d) Planck constant

Answer: (c) Equilibrium constant — For the formation of a complex ion.

77. Which is omitted from the K_{stab} expression?

(a) H_2O (b) The ligand
(c) The complex (d) The metal ion

Answer: (a) H_2O — Water is the solvent.

78. $\log_{10}K_{stab}$ of $[CuCl_4]^{2-}$ is:

(a) 18.8 (b) 5.6
(c) 13.1 (d) 1.0

Answer: (b) 5.6 — Table 18.5.

79. $\log_{10}K_{stab}$ of $[Cu(EDTA)]^{2-}$ is:

(a) 5.6 (b) 8.8
(c) 13.1 (d) 18.8

Answer: (d) 18.8 — The most stable of the three.

studysapiens.com

80. The extra stability of complexes with polydentate ligands is called the:

(a) Inductive effect (b) Shielding effect
(c) Chelating effect (d) Photoelectric effect

Answer: (c) Chelating effect — Section 18.8.

81. A higher K_{stab} means the complex is:

(a) More stable (b) Less stable
(c) Coloured (d) Insoluble

Answer: (a) More stable — Equilibrium lies far to the right.

82. Massive dilution of a stable complex shifts the equilibrium:

(a) To the right (b) To the left
(c) Nowhere (d) To the precipitate

Answer: (b) To the left — The hydrated ion re-forms.

18.9 STEREOISOMERISM

83. Cis-trans isomerism is common in:

(a) Linear and tetrahedral complexes
(b) Square planar and octahedral complexes
(c) Linear complexes only
(d) Tetrahedral complexes only

Answer: (b) Square planar and octahedral complexes — Section 18.9.1.

84. In a cis isomer the identical ligands are at an angle of:

(a) 90° (b) 180°
(c) 109.5° (d) 120°

Answer: (a) 90° — Adjacent positions.

85. Cisplatin is used as:

(a) An antacid (b) A catalyst
(c) An anti-cancer drug (d) A pigment

Answer: (c) An anti-cancer drug — Transplatin is not.

86. Transplatin is:

(a) Polar (b) Tetrahedral
(c) Optically active (d) Non-polar

Answer: (d) Non-polar — Its bond dipoles cancel.

87. Activated cisplatin binds to which base of DNA?

(a) Adenine (b) Guanine
(c) Thymine (d) Cytosine

Answer: (b) Guanine — It creates cross-links.

88. Optical isomers are also called:

(a) Anomers (b) Allotropes
(c) Tautomers (d) Enantiomers

Answer: (d) Enantiomers — Non-superimposable mirror images.

89. $[Ni(en)_3]^{2+}$ shows:

(a) Cis-trans isomerism (b) No isomerism
(c) Optical isomerism (d) Chain isomerism

Answer: (c) Optical isomerism — Octahedral with bidentate ligands.

90. An isomer that rotates plane-polarized light to the left is:

(a) Laevorotatory (b) Dextrorotatory
(c) Achiral (d) Trans

Answer: (a) Laevorotatory — Shown by "–".

91. $[Co(NH_3)_4(H_2O)_2]^{3+}$ is a complex of the type:

(a) $[MA_2B_2]$
(b) $[MA_6]$
(c) $[MA_4B_2]$
(d) $[MA_3B_3]$

Answer: (c) $[MA_4B_2]$ — It shows cis-trans isomerism.

18.10 REDOX REACTIONS

92. A redox reaction is feasible when $E°_{cell}$ is:

(a) Negative (b) Less than –1 V
(c) Zero (d) Positive

Answer: (d) Positive — Section 18.10.

93. The species with the more positive E° value is:

(a) Oxidized (b) Reduced
(c) Unchanged (d) Precipitated

Answer: (b) Reduced — It gains electrons.

94. E° for MnO_4^-/Mn^{2+} is:

(a) +0.77 V (b) +0.54 V
(c) +1.52 V (d) +0.15 V

Answer: (c) +1.52 V — Section 18.10.1.

StudySapiens.com

95. In acid solution MnO_4^- is reduced to:

(a) Mn^{2+}
(b) MnO_2
(c) MnO_4^{2-}
(d) Mn

Answer: (a) Mn^{2+} — It gains 5 electrons.

96. The ethanedioate ion is oxidized to:

(a) CO
(b) C
(c) CO_3^{2-}
(d) CO_2

Answer: (d) CO_2 — It is the reducing agent.

97. $E°_{cell}$ for the $MnO_4^-/C_2O_4^{2-}$ reaction is:

(a) +2.01 V
(b) –1.03 V
(c) +1.03 V
(d) +0.49 V

Answer: (c) +1.03 V — 1.52 – 0.49.

98. How many electrons does one MnO_4^- ion gain?

(a) 2
(b) 5
(c) 7
(d) 1

Answer: (b) 5 — Mn goes from +7 to +2.

99. In the reaction with MnO_4^-, Fe^{2+} acts as:

(a) Reducing agent
(b) Oxidizing agent
(c) Catalyst
(d) Ligand

Answer: (a) Reducing agent — It is oxidized to Fe^{3+}.

100. $E°$ for Fe^{3+}/Fe^{2+} is:

(a) +1.52 V
(b) +0.15 V
(c) +0.77 V
(d) +0.54 V

Answer: (c) +0.77 V — Section 18.10.2.

101. The precipitate formed when excess KI is added to copper(II) sulphate is:

(a) CuI
(b) CuI_2
(c) Cu_2O
(d) Cu

Answer: (a) CuI — A white precipitate.

102. The Cu^{2+}/I^- reaction goes forward because:

(a) $E°$ is positive
(b) Iodine is a gas
(c) It is heated
(d) CuI precipitates out

Answer: (d) CuI precipitates out — Le Chatelier’s principle.

study-sapiens.com

103. Iodine formed in this reaction is estimated by titration with:

(a) Permanganate (b) Thiosulphate
(c) Oxalate (d) EDTA

Answer: (b) Thiosulphate — Section 18.10.3.

104. In a redox titration with potassium permanganate, the indicator is:

(a) Starch (b) Phenolphthalein
(c) The permanganate itself (d) Methyl orange

Answer: (c) The permanganate itself — Its purple colour is discharged; it is self-indicating.

105. The indicator used in iodine–thiosulphate titrations is:

(a) Methyl orange (b) Phenolphthalein
(c) Litmus (d) Starch

Answer: (d) Starch — Starch gives a blue-black colour with iodine.

106. The colour of the permanganate ion in solution is:

(a) Green (b) Purple
(c) Orange (d) Blue

Answer: (b) Purple — It becomes almost colourless Mn^{2+} on reduction in acid.

107. The oxidation state of manganese in the permanganate ion is:

(a) +7 (b) +4
(c) +6 (d) +2

Answer: (a) +7 — Four oxygens at −2 with an overall charge of −1.

studysapiens.com